

STAT 525

Chapter 1

Linear Regression with One Predictor

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Goals of Regression Analysis

- Serve three purposes
 - Describes an association between X and Y
 - * In some applications, the choice of which variable is X and which is Y can be arbitrary
 - * Association generally does not imply causality
 - In experimental settings, helps select X to control Y at the desired level
 - Predict a future value of Y at a specific value of X
- **Always** need to consider scope of the model *Range*
 - The specific model may only show the association within a certain range.

Example: Leaning Tower of Pisa

- Annual measurements of its lean available
- Measured in tenths of a mm > 2.9 meters
- Prior to recent repairs, its lean was increasing over time
- Goals:
 - To **characterize** ^{y} lean over ^{x} time
 - To **predict** future observations

The Data Set

Index X Y
Obs year lean

1	1975	642
2	76	644
3	77	656
4	78	667
5	79	673
6	80	688
7	81	696
8	82	698
9	83	713
10	84	717
11	85	725
12	86	742
13	1987	757

Data taken from Exercise 10.8, p698 in Moore and McCabe,
Intro to the Practice of Statistics, 3rd ed.

The Data and Relationship

- Response/Dependent variable: lean (Y)
- Explanatory/Independent variable: year (X)
- Observe lean from 1975 - 1987
- Is there a relationship between Y and X ?

◦ Predict future relationship

Ex:) $X = \overset{\text{year}}{2000}$?

To Generate a Scatterplot in SAS

```
DATA a1; INPUT year lean @@;
```

name *x* *y* *tells SAS to read multiple observations from the same line in 'CARDS'*

CARDS; *→ 'Manually input.'*

```
75 642 76 644 77 656 78 667 79 673 80 688
81 696 82 698 83 713 84 717 85 725 86 742
87 757 102
```

Missing
Year = 2000 *Placeholder*

```
PROC PRINT DATA=a1; WHERE lean NE .; RUN;
```

(filters out) without missing included

```
SYMBOL1 V=CIRCLE I=SM70;
```

Shape of dots to be circular *smooth level = 70*

```
PROC GLOT DATA=a1;
```

Joint

```
    PLOT lean*year / FRAME; WHERE lean NE .;
```

Y x X

```
RUN;
```

boxed frame around the axis

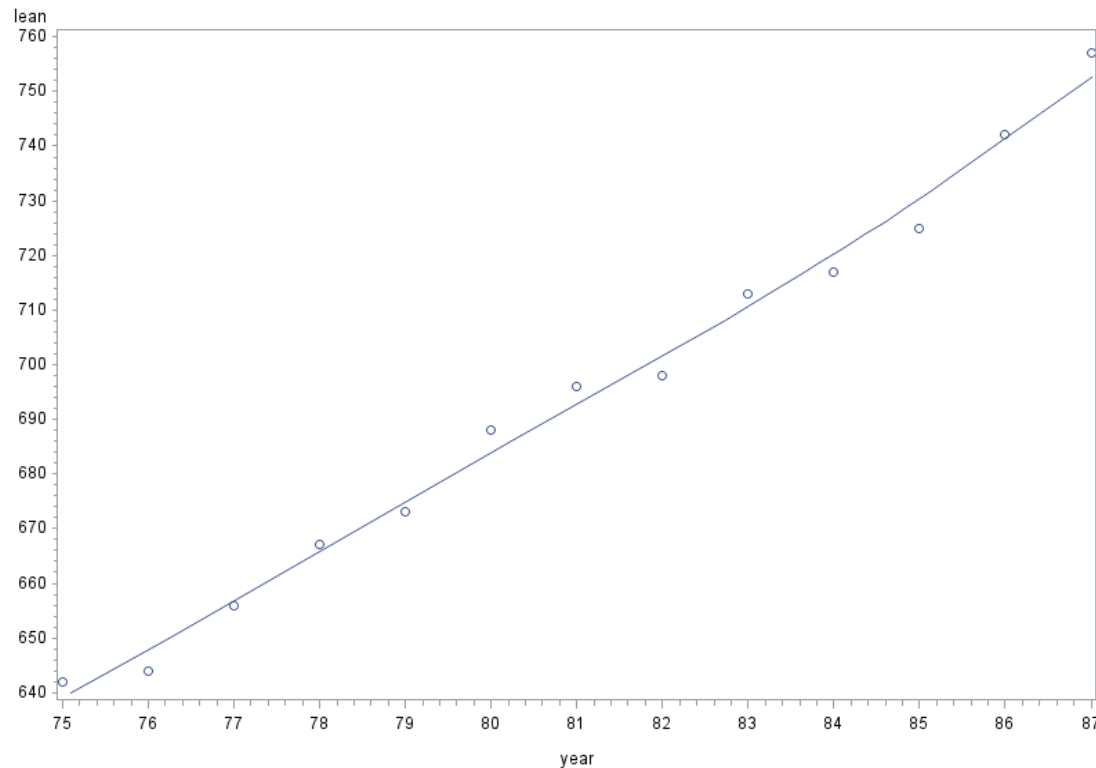
Comments:

@@: double trailing @ allows creating multiple observations from a single record.

SM<70><S>: plot smooth curve with smoothness level 70 (0-99). *“Connecting the dots”*

What is the Trend?

- Should always plot the data first! Visualization
- A first look at how Y changes as X is varied is available from a scatterplot.
- Helps to visually detect abnormalities in data set.



Linear Trend?

'Regression Linear'

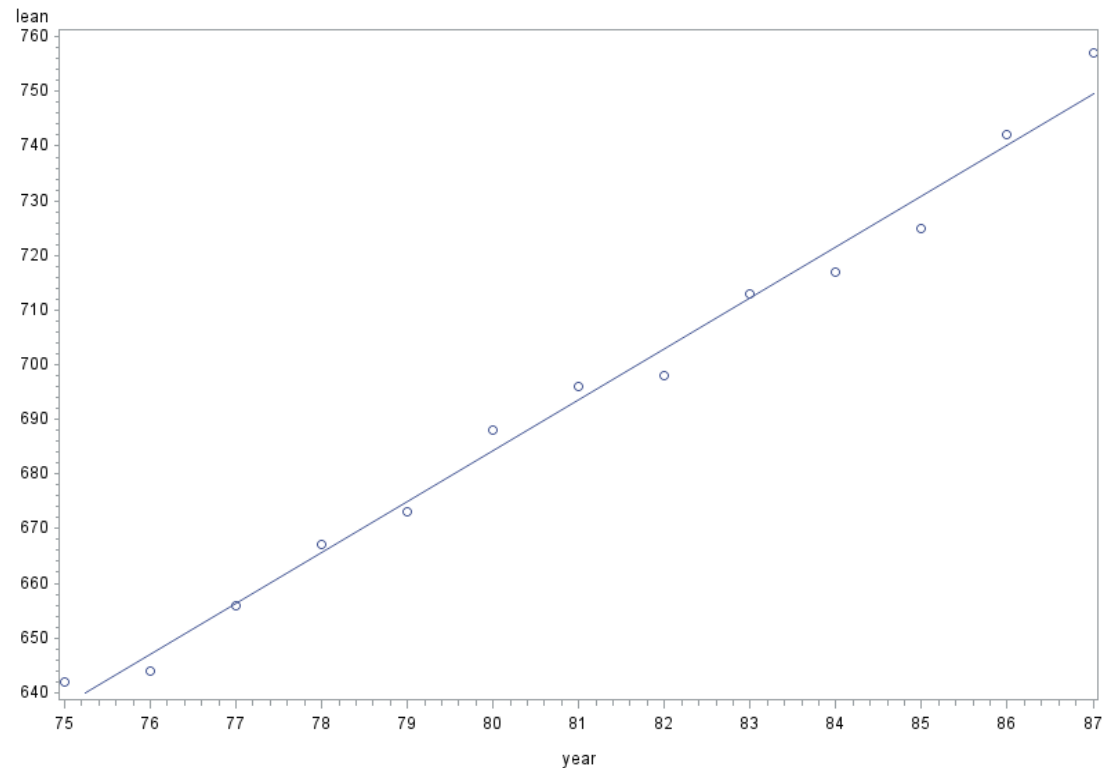
```
SYMBOL1 V=CIRCLE I=r1;
```

```
PROC GPLOT DATA=a1;
```

```
    PLOT lean*year / FRAME; WHERE lean NE .;
```

```
RUN;
```

R<L/Q/C>: plot a Linear Regression result.



Straight Line Equation

- Straight line describes “^{‘estimation’}curve” well
- Formula for a straight line

$$E(Y_i) = \beta_0 + \beta_1 X_i, \text{ or } E(Y_i|X_i) = \beta_0 + \beta_1 X_i$$

- β_0 is the intercept
- β_1 is the slope
- Need to **estimate** β_0 and β_1
i.e. determine their plausible values from the data
- Will use method of **least squares** (OLS estimator).

Simple Linear Regression Model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- β_0 is the intercept
- β_1 is the slope
- ε_i is the i^{th} random error term Error around the center of distribution:
 - Mean 0, i.e. $E(\varepsilon_i) = 0$ no systematic bias
 - Constant Variance σ^2 , i.e. $Var(\varepsilon_i) = \sigma^2$ Homoscedasticity
 - Uncorrelated, i.e. $Cov(\varepsilon_i, \varepsilon_j) = 0$
 - Independent to X_i if X_i is random

$$Z = E Z + \underbrace{(Z - E Z)}_{\text{Error}}$$

Features of the Model

- $Y_i =$ deterministic term $(\beta_0 + \beta_1 X_i)$ + random term (ε_i)
- Implies Y_i is a random variable (for both fixed or random X_i cases)
 - $E(Y_i) = \beta_0 + \beta_1 X_i + 0$, or
 $E(Y_i|X_i = x) = \beta_0 + \beta_1 x + 0$ (underlying relationship)
 - $Var(Y_i) = 0 + \sigma^2$, or
 $Var(Y_i|X_i = x) \equiv \sigma^2$ (constant variance)
 - $Cov(Y_i, Y_j) = Cov(\varepsilon_i, \varepsilon_j) = 0$, or
 $Cov(Y_i, Y_j|X_i = x, X_j = x') \equiv 0$. *implies that errors are uncorrelated*
- For simplicity, we assume X_i 's are fixed unless specified otherwise.

- Least Square • We want to find the "best" straight line
 • Choose the line that makes the vertical distance between Y_i and $aX_i + b$ as small as possible.
- point line

Estimation of Regression Function

"Finding a line closest to the dataset"

- Consider the deviation of observed data Y_i from a straight line with slope a and intercept b ,



point — line

$$Y_i - (aX_i + b)$$

it measures how good the line $ax + b$ fits the data (X_i, Y_i) in terms of vertical distance

- Method of least squares (smallest sum of squared derivation)

– Find the value of a and b which minimize

$$Q = \sum_{i=1}^n [Y_i - (aX_i + b)]^2$$

② → • Avoid canceling out positive/negative errors
 • Penalizes large deviations more heavily.

– Motivated by $E(Y) = \arg \min_b E(Y - b)^2 \approx \arg \min_b \sum (Y_i - b)^2 / n$.

Mean of Y : $E(Y) = b$

Shortest Expected Distance

argument of
the minimum

Estimating/Interpreting the Slope

- β_1 is the true unknown slope
 - Defines change in $E(Y)$ for change in X , i.e.,

$$\beta_1 = \frac{\overset{\text{change}}{\Delta E(Y)}}{\Delta X}$$

- b_1 is the least squares estimate of β_1

$\hat{\beta}_1$

$$b_1 = \frac{\overset{\text{Covariance of } X \text{ and } Y}{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}}{\underset{\text{variance of } X}{\sum_{i=1}^n (X_i - \bar{X})^2}}$$

- When will b_1 be negative or positive?

$b_1 > 0$ positive association

$b_1 < 0$ negative association

Depends on numerator $\sum (x_i - \bar{x})(y_i - \bar{y})$
(ie. Covariance + / -)

Estimating/Interpreting the Intercept

- β_0 is the true unknown intercept

- β_0 is the expected value of Y under $X = 0$

$$E(Y) = \beta_1 X + \beta_0 = \cancel{\beta_1 \times X} + \beta_0 = \beta_0$$

- Sometimes not of interest (scope of model)

- b_0 is the least squares estimate of β_0

$\hat{\beta}_0$

$$b_0 = \bar{Y} - b_1 \bar{X}$$

that is, the fitted line goes through $(\bar{X}, \bar{Y})$.

Properties of Estimates

- Under the *Gauss-Markov* theorem, these least squares estimators
 - are ^{b_0, b_1} **unbiased**, $E(b_i) = \beta_i$
 - Have **minimum variance** among all unbiased linear estimators
- In other words, these estimates are the most precise among all estimators which satisfy
 - b_i is of form $\sum k_i Y_i$
 - $E(b_i) = \beta_i$

Estimated Regression Line

Used for:

- Predict values
- Calculate residuals
- Evaluate model fit

- Using the estimated parameters, the fitted regression line is

$$\hat{Y}_i = b_0 + b_1 X_i$$

where $\hat{Y}_i$ is the estimated value at X_i (Fitted value).

- Fitted value $\hat{Y}_i$ is also an estimate of the mean response $E(Y_i)$
- Extension of the Gauss-Markov theorem ↓ not individual outcomes
 - $E(\hat{Y}_i) = E(Y_i)$
 - $\hat{Y}_i$ has minimum variance among all unbiased linear estimators

Example: Leaning Tower of Pisa

Based on the following table

1. Obtain the least squares estimate of β_0 and β_1 .
2. State the regression function
3. Obtain a point estimate for the year 2002 ($X = 102$)
4. State the expected change in lean over two years

		Covariance of X and Y				
	X	Y	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X})(Y - \bar{Y})$	$(X - \bar{X})^2$
	75	642	-6	-51.6923	310.1538	36
	76	644	-5	-49.6923	248.4615	25
	77	656	-4	-37.6923	150.7692	16
	78	667	-3	-26.6923	80.0769	9
	79	673	-2	-20.6923	41.3846	4
	80	688	-1	-5.6923	5.6923	1
	81	696	0	2.3077	0	0
	82	698	1	4.3077	4.3077	1
	83	713	2	19.3077	38.6154	4
	84	717	3	23.3077	69.9231	9
	85	725	4	31.3077	125.2308	16
	86	742	5	48.3077	241.5385	25
	87	757	6	63.3077	379.8462	36
Σ	1053	9018	0	0	1696	182

$$n=13$$

Answer

1. Obtain the least squares estimate of β_0 and β_1 .

$$b_1 = \frac{1696}{182} = 9.3187, \quad b_0 = \left\{ \frac{9018}{13} - 9.3187 \right\} \frac{1053}{13} = -61.1224$$

Handwritten annotations: $(x-\bar{x})(y-\bar{y})$ above 1696, $(x-\bar{x})^2$ below 182, $\bar{y}$ above 9018, n below 13, b_1 above 9.3187, $\bar{x}$ above 1053, n below 13.

- ✓ 2. State the regression function

$$\hat{Y}_i = -61.1224 + 9.3187X_i$$

- ✓ 3. Obtain a point estimate for the year 2002 ($X = 102$)

$$(\hat{Y}|X = 102) = -61.1224 + 9.3187(102) = 889.3850$$

- ✓ 4. State the expected change in lean over two years :

Since the slope is 9.3187, a two unit increase in X results in a $2 \times 9.3187 = 18.6374$ increase in lean.

Handwritten annotation: 1 unit ↑ in year above 9.3187

Residuals

Defined after line is chosen

- The **residual** is the difference between the **observed and fitted values**

$$e_i = Y_i - \hat{Y}_i$$

real -

values predicted by
the "best" line we chose

- This is not the error term $\varepsilon_i = Y_i - E(Y_i)$
- The **e_i is observable** while ε_i is not
- Residuals are highly useful in assessing the appropriateness of the model

Properties of Residuals

- $\sum e_i = 0$
- $\sum e_i^2$ are minimized
- $\sum Y_i = \sum \hat{Y}_i$
- $\sum X_i e_i = 0$
- $\sum \hat{Y}_i e_i = 0$

These properties follow directly from the least squares criterion and normal equations (pg 23-24)

Restricted Maximum Likelihood

ReML Estimation of Error Variance

- In single population (i.e., ignoring X)

$$s^2 = \frac{\sum (Y_i - \bar{Y})^2}{n - 1}$$

- unbiased estimation *Estimating variance around the mean*
- One df lost by using $\bar{Y}$ in place of μ_Y

- In regression model

$$s^2 = \frac{\sum (Y_i - \hat{Y}_i)^2}{n - 2} \xrightarrow{\text{Numerator:}} (\text{Residuals})^2 = \frac{SSE}{n-2} = \hat{\sigma}_{LSE}^2$$

- unbiased estimation *Estimating variance around fitted regression line*
- Two df lost by using (b_0, b_1) in place of (β_0, β_1) *dof = n-2*
- Also known as the **mean square error (MSE)**

biased estimation: doesn't adjust for dof lost from estimating β_0 and β_1

$$\hat{\sigma}_{MLE}^2 = \frac{SSE}{n}$$

PROC REG in SAS: Leaning Tower of Pisa

```
PROC REG DATA=a1;
  MODEL lean=year / confidence interval for 'Beta' CLB P R;
  OUTPUT OUT=a2 P=pred R=resid;
  ID year;
RUN;
```

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	15804	15804	904.12	<.0001
Error	11	192.28571	17.48052		
Corrected Total	12	15997			

Root MSE	4.18097	R-Square	0.9880
Dependent Mean	693.69231	Adj R-Sq	0.9869
Coeff Var	0.60271		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	95% Confidence Limits
β_0 Intercept	1	-61.12088	25.12982	-2.43	0.0333	-116.43124 -5.81052
β_1 year	1	9.31868	0.30991	30.07	<.0001	8.63656 10.00080

		Output Statistics				
		Dependent	Predicted	Std Error		Std Error
Obs	year	Variable	Value	Mean Predict	Residual	Residual
1	75	642.0000	637.7802	2.1914	4.2198	3.561
2	76	644.0000	647.0989	1.9354	-3.0989	3.706
3	77	656.0000	656.4176	1.6975	-0.4176	3.821
4	78	667.0000	665.7363	1.4863	1.2637	3.908
5	79	673.0000	675.0549	1.3149	-2.0549	3.969
6	80	688.0000	684.3736	1.2003	3.6264	4.005
7	81	696.0000	693.6923	1.1596	2.3077	4.017
8	82	698.0000	703.0110	1.2003	-5.0110	4.005
9	83	713.0000	712.3297	1.3149	0.6703	3.969
10	84	717.0000	721.6484	1.4863	-4.6484	3.908
11	85	725.0000	730.9670	1.6975	-5.9670	3.821
12	86	742.0000	740.2857	1.9354	1.7143	3.706
13	87	757.0000	749.6044	2.1914	7.3956	3.561
14	102	.	889.3846	6.6107	.	.

Comments:

CLB: confidence intervals for the β estimations

P R: display predicted/fitted values and residual values

output: Output statistics into a new data set

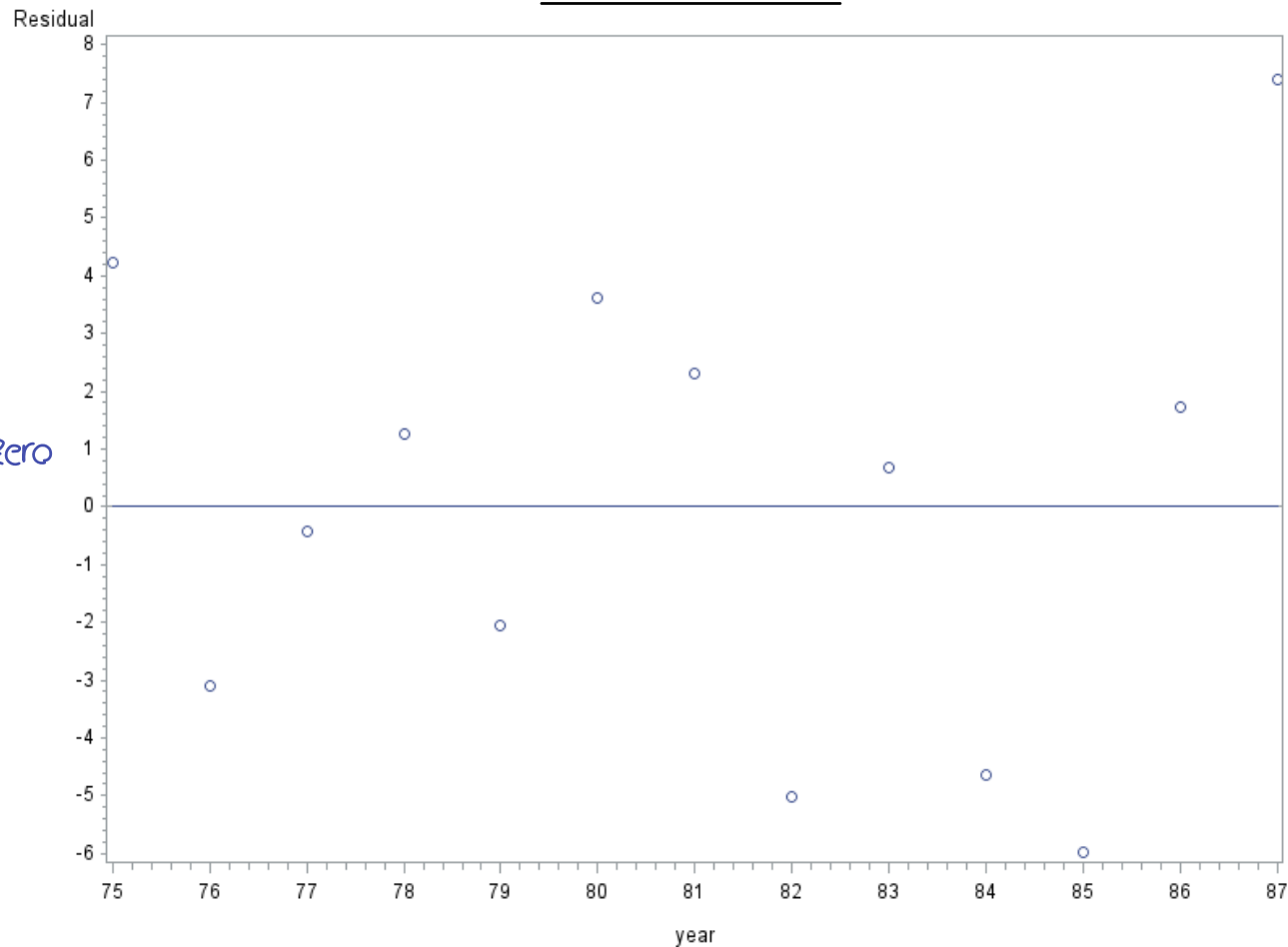

```
PROC Gplot DATA=a2;  
PLOT resid*year / FRAME VREF=0;  
WHERE lean NE .;  
RUN;
```

Detect?

- nonlinearity
- heteroscedasticity
- outliers ...

Residual Plot

✓ good!
Scatter around zero



Normal Error Regression Model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \varepsilon_i \sim^{iid} N(0, \sigma^2)$$

- β_0 is the intercept
- β_1 is the slope
- the random error term is assumed to be **independent normally** distributed
- Defines distribution of random variable Y_i

$$Y_i \sim N(\beta_0 + \beta_1 X_i, \sigma^2)$$

Comments

normality isn't required for unbiased estimates

- The **normality assumption doesn't affect** the **validity** and **unbiasedness** of the least square estimators
- The normality assumption will greatly simplify the theory of analysis beyond estimations
- The normality assumption makes it easy to construct confidence intervals / perform hypothesis tests
- Most inferences are only sensitive to large departures from normality
- See pages 26-27 for more details

Most methods are robust unless normality is severely violated.

Comments

- Assumption of normality gives us more choices of methods for parameter estimation

$$\begin{aligned} f_i &= \text{the likelihood of } Y_i \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} (Y_i - \beta_0 - \beta_1 X_i)^2 \right\} \end{aligned}$$

- Likelihood function $L = \prod f_i$ (i.e. the joint probability distribution of the observations, viewed as function of parameters)
- Find β_0 , β_1 and σ^2 which maximizes L .
- Obtain similar estimators b_0 and b_1 for β_0 and β_1 , but slightly different estimators for σ^2 (see HW#1)

↓
MLE tends to slightly underestimate
unless adjusted.

$$\hat{\sigma}_{MLE}^2 = \frac{SSE}{n}$$

↓
denominator bigger

$$\hat{\sigma}_{LSE}^2 = \frac{SSE}{n-2}$$

Chapter Review

- Description of Linear Regression Model
- Least Squares & Parameter Estimation
- Fitted Regression Line
- Normality Assumption
- PROC REG in SAS: First Touch